

Quantum Cryptanalytic Algorithms and the Transition to Post-Quantum Cryptography: A Survey of Emerging Threats and NIST PQC Standards

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Abstract

The rapid advancement of quantum computing presents a fundamental challenge to classical cryptographic systems, necessitating a comprehensive understanding of both algorithmic and hardware progress. This survey examines the evolving quantum threat landscape along three critical dimensions. First, it reviews the evolution of quantum cryptanalytic algorithms, from foundational results such as Shor’s algorithm and Grover’s algorithm to more recent refinements and alternative constructions, including Regev-type frameworks, highlighting how these developments affect both asymmetric cryptosystems and the effective security of symmetric primitives. Second, it analyses recent reductions in quantum resource requirements, with particular emphasis on fault-tolerant architectures and quantum error-correction techniques that significantly lower physical qubit overhead under optimistic assumptions. Third, it evaluates progress in quantum hardware, including improvements in qubit coherence, noise resilience, and material platforms, which collectively influence the feasibility and timeline of large-scale fault-tolerant quantum computation.

In parallel, the paper reviews the status of post-quantum cryptographic standardisation, focusing on National Institute of Standards and Technology FIPS 203–205 and ongoing developments, and discusses implementation-level vulnerabilities to highlight deployment challenges. A risk assessment framework based on Mosca’s inequality is applied to contextualise these advances under the “harvest now, decrypt later” threat model. The analysis demonstrates that cryptographic risk is shaped not only by algorithmic breakthroughs but also by reductions in qubit overhead and improvements in hardware stability, both of which compress the timeline to cryptanalytically relevant quantum capabilities. Based on these findings, the paper provides practical recommendations to support cryptographic agility and a secure transition to post-quantum standards.

Keywords— Cryptanalytically Relevant Quantum Computer, Post-Quantum Cryptography, Shor’s Algorithm, Grover’s Algorithm, Mosca’s inequality, NIST FIPS 203–205.

1 Introduction

Modern digital security is foundational to global communication and commerce, relying upon a combination of asymmetric and symmetric cryptographic primitives. Asymmetric systems, such as RSA and Elliptic Curve Cryptography (ECC), facilitate essential services including secure key exchange, digital signatures, and identity verification. These schemes derive their security from the computational hardness of integer factorization and the discrete logarithm problem (DLP)—mathematical challenges that are efficiently solvable via Shor’s quantum algorithm. Consequently, classical public-key infrastructures are entirely vulnerable to the eventual realization of a cryptanalytically relevant quantum computer (CRQC; also referred to as a cryptographically relevant quantum computer). [1, 9, 28] Symmetric primitives, including AES and

SHA-2, are not broken outright by quantum algorithms; however, Grover’s algorithm provides a generic quadratic speed-up for brute-force search and thus reduces their effective security margins, motivating the use of larger key and digest sizes (e.g., AES-256, SHA-384/512) for post-quantum robustness. [2, 4, 6]

The timeline for these threats is increasingly shaped by a global “fault-tolerance race,” characterized by divergent architectural approaches. Google and IBM have primarily focused on superconducting transmon qubits, while Microsoft pursues a qualitatively different path based on topological qubits, with recent claims of a quantum-computing breakthrough, albeit accompanied by scientific skepticism.[13, 14] Recent peer-reviewed theoretical approaches, including logical operator gauging, suggest the potential for quantum LDPC (qLDPC) codes to reduce error-correction overhead. In parallel, experimental efforts such as IBM’s prototype processors explore feasible hardware architectures; however, scalable fault-tolerant performance has not yet been demonstrated.[11, 12]

In response, NIST released the first generation of standardized post-quantum algorithms (FIPS 203/204/205) and announced a forthcoming signature standard (FIPS 206) to strengthen long-term security postures.[17, 18, 19, 21]

2 Problem Statement

A CRQC would break widely deployed public-key systems based on integer factorization and discrete logarithms, endangering confidentiality and authenticity of internet-scale systems.[26, 27, 28] Even before such a device exists, adversaries can mount *harvest now decrypt later* (HNDL) attacks by storing encrypted traffic today for future decryption—elevating the urgency of PQC migration for data with long confidentiality lifetimes.[25]

3 Objectives

The purpose of this survey is to synthesize current research on quantum cryptanalysis, evaluate the vulnerability of classical cryptographic systems to quantum attacks, and identify practical guidance for post-quantum transition. Accordingly, this paper aims to achieve the following objectives:

- Identify the quantum algorithms that most critically threaten contemporary cryptosystems and summarize current resource estimates.
- Review the status, standardization progress, and technical properties of NIST-standardized and emerging post-quantum cryptographic (PQC) schemes.
- Analyze the practical implications of Shor’s and Grover’s algorithms for asymmetric cryptography, symmetric encryption, and cryptographic hashing, and discuss recommended security parameter adjustments.
- Highlight recent implementation-level attacks (e.g., KyberSlash) and extract lessons for secure PQC deployment.
- Provide migration recommendations and a future research outlook grounded in current standardization guidance and threat projections.
- Examine the applicability and limitations of Mosca’s Inequality as a framework for post-quantum risk assessment, and discuss potential extensions toward continuous HNDL-oriented risk models incorporating temporal exposure accumulation and information-value persistence.

4 Literature Review

4.1 Shor-type Factoring and Resource Estimates

Shor’s algorithm[1] provides a polynomial-time quantum method for factoring and discrete logarithms, with asymptotic complexity on the order of $\tilde{O}(k^2)$ to $\tilde{O}(k^3)$ (with optimized arithmetic) for a k -bit modulus. Under optimistic assumptions about fault-tolerant quantum hardware—particularly logical gate speeds in the MHz range and efficient magic-state distillation—these scalings suggest that factoring an RSA-2048 modulus could, in principle, be achieved in hours to days. Concrete resource estimates, however, remain highly sensitive to error correction schemes and architectural constraints.

Early surface-code analyses (e.g., by Craig Gidney and Martin Ekerå) indicated requirements on the order of 20 million physical qubits for sub-day runtimes. More recent refinements, incorporating improved compilation and distillation strategies, reduce this to below one million physical qubits under favorable assumptions, with runtimes ranging from hours to under a week.[7]

Alternative theoretical approaches based on QLDPC codes aim to substantially reduce qubit overhead. For example, the “Pinnacle” architecture[8] proposes a fault-tolerant design in which RSA-2048 factoring could be achieved with fewer than $\sim 10^5$ physical qubits under specific hardware assumptions, including physical error rates on the order of 10^{-3} and microsecond-scale error-correction cycles.

However, these estimates rely on tightly integrated architectural features—such as high-throughput magic state distillation (“magic engines”), modular QLDPC processing units, and efficient classical feedforward—that are not yet experimentally demonstrated at scale. As a result, while such proposals indicate a plausible path to order-of-magnitude reductions in qubit counts relative to surface-code-based designs, they remain significantly more speculative and hardware-sensitive, often involving more demanding implementation requirements and potentially longer runtimes.

Overall, while the asymptotic advantage of Shor’s algorithm is well established, practical timelines for breaking RSA-2048 depend critically on advances in fault-tolerant architectures, error correction, and large-scale quantum hardware.

4.2 Regev’s Efficient Quantum Factoring and Elliptic Curves

Regev introduced a factoring framework with a lower gate complexity per run but more repetitions under a heuristic number-theoretic assumption, with analyses exploring space/noise trade-offs and practical considerations.[9] Subsequently, Barbulescu, Barcau, and Paşol extended this framework to elliptic-curve discrete logarithms (ECDLP), identifying parameter regimes where it may outperform Shor asymptotically and describing instances of failure/success under specific heuristics.[10]

4.3 Grover’s Algorithm and Symmetric Primitives

Grover’s algorithm yields a quadratic speed-up for unstructured search; in cryptography this implies that idealized key search over a k -bit key space requires roughly $2^{k/2}$ evaluations.[2, 23] This reduction affects all symmetric primitives, including block ciphers, MACs, and hash functions. However, unlike Shor’s algorithm for RSA and ECC—which completely breaks those schemes—a quadratic speed-up merely reduces effective security, rather than eliminating it outright.

While no known algorithm asymptotically surpasses Grover’s bound for unstructured symmetric-key search, ongoing advances in hybrid quantum optimization, oracle engineering, circuit reduction, and variational quantum algorithms may progressively reduce the practical resource overheads associated with future quantum computational and cryptanalytic implementations.[5]

4.4 NIST Guidance on Symmetric Security in the Quantum Era

NIST key-management recommendations and cost analyses provide baseline guidance for selecting symmetric-key parameters suitable for long-lived data in the presence of quantum adversaries, including the use of AES-256 and SHA-384/512 to retain adequate post-quantum security margins.[24]

While existing literature extensively characterizes the algorithmic capabilities of quantum adversaries, the practical realization of such threats is fundamentally constrained by the underlying quantum hardware. Accordingly, a system-level understanding of current quantum computing platforms is necessary to contextualize cryptanalytic feasibility.

5 Quantum Hardware Landscape and Vendor Comparison

The realization of a CRQC is constrained not only by algorithmic advances but also by the physical characteristics of qubit implementations. As of 2026, no dominant hardware paradigm has emerged; instead, multiple competing qubit modalities coexist, each with distinct trade-offs in coherence, temperature requirements, scalability, and error susceptibility. These hardware-level considerations play a decisive role in determining the feasibility and timeline of large-scale quantum computation.

5.1 Major Industrial Approaches

Leading technology vendors have adopted fundamentally different qubit strategies[38, 39, 40]:

- **IBM / Google:** Superconducting transmon qubits based on Josephson junction circuits.
- **Intel:** Silicon spin qubits implemented in semiconductor quantum dots.
- **Microsoft:** Topological qubits based on Majorana quasiparticles (currently experimental).

These approaches differ in physical realization, which directly influences operational constraints, coherence properties, and scalability pathways.

5.2 Comparative Hardware Characteristics

Table 1: Comparison of major quantum computing hardware platforms as of 2026. Reported values are representative approximate estimates and may vary across implementations and experimental conditions.

Attribute	IBM / Google	Intel	Microsoft
Qubit Type	Transmon	Spin (Si)	Topological
Physical Platform	Superconducting circuits	Quantum dots	Majorana modes
Operating Temperature	~10–20 mK	~50–100 mK	~20 mK (target)
Coherence Time	~50–200 μ s	μ s–ms	Theoretical (long)
Gate Speed	~10–100 ns	ns– μ s	No validated value
Scalability	100–1000+ qubits	Early-stage	Experimental
Key Limitation	Decoherence, cooling	Variability	Unproven physics

As shown in Table 1, superconducting platforms currently dominate in scale and maturity, while silicon spin qubits offer improved thermal tolerance and potential compatibility with existing semiconductor manufacturing. Topological approaches aim to address decoherence at a

fundamental level but remain experimentally unproven. Consequently, hardware modality—not merely qubit count—plays a central role in assessing CRQC feasibility.

5.3 Emerging Hardware Trends

Recent research reveals several converging trends across quantum hardware platforms[41, 42]:

5.3.1 Relaxation of Cryogenic Constraints

Traditional superconducting systems require ultra-low temperatures ($\sim 10\text{--}20$ mK). Emerging qubit platforms demonstrate operation at elevated temperatures (≥ 100 mK), reducing reliance on extreme cryogenic infrastructure and simplifying system engineering.

5.3.2 Noise-Resilient Qubit Design

The field is transitioning from minimizing environmental noise to engineering systems that are intrinsically robust to noise. This includes advances in materials, fabrication processes, and qubit architectures designed to tolerate non-ideal operating conditions.

5.3.3 Material-Driven Quantum Computing

Materials science is increasingly central to quantum computing progress. The choice of substrate directly influences coherence time, noise susceptibility, and scalability. Advances in superconducting materials, silicon purification, and alternative hosts are driving improvements across platforms.

5.3.4 Coherence–Scalability Trade-off

A persistent trade-off exists between coherence and scalability. Superconducting systems are highly scalable but exhibit relatively short coherence times, whereas spin-based and atomic systems often achieve longer coherence but present challenges in large-scale integration.

5.4 Impact of Recent Advances

Recent work on electron qubits hosted in solid neon introduces a new design axis characterized by higher operating temperatures ($\sim 100\text{--}400$ mK) and improved noise resilience.[43] While coherence times remain on the order of microseconds, these systems demonstrate that qubit stability may be achieved under less restrictive thermal conditions.

5.4.1 Implications

- **Reduced Thermal Constraints:** Higher operating temperatures alleviate the engineering burden associated with dilution refrigeration.
- **Alternative Scaling Pathways:** Novel material platforms suggest that future quantum systems may not rely exclusively on superconducting architectures.
- **Increased Timeline Uncertainty:** Hardware breakthroughs introduce variability in projections of CRQC realization timelines.

5.5 Relevance to Cryptanalytic Risk

The diversity of hardware approaches introduces structural uncertainty into quantum threat modelling. Progress is inherently non-linear and architecture-dependent, with potential breakthroughs arising from unexpected physical implementations. Hardware advances may reduce error-correction overhead, increase feasible qubit counts, and shorten execution timelines.

Accordingly, the timeline to cryptographically relevant quantum computation is not solely a function of algorithmic efficiency, but is increasingly determined by advances in materials science and quantum hardware engineering.

6 Analysis of Quantum Cryptanalytic Algorithms

6.1 RSA and Finite-Field DLP

- **Shor + surface-code baselines.** RSA-2048 factoring feasibility was historically tied to multi-million-qubit machines; more recent analyses compress qubit counts by trading space for time and optimizing non-Clifford resources.[7]
- **Under one million qubits.** Sub-million-qubit executions with week-scale runtimes become plausible under optimistic hardware parameters and advanced error-correction strategies.[7]
- **QLDPC pathways.** Emerging theoretical models, such as the proposed Pinnacle Architecture, suggest that the physical qubit requirement for RSA-2048 factorization could be reduced to approximately 10^5 qubits with a month-scale runtime. However, these estimates remain contingent on the realization of high-speed classical decoding and non-local qubit connectivity—architectural features that have yet to be demonstrated at a cryptanalytic scale.[8]

6.2 Elliptic-Curve Cryptography

Shor’s algorithm breaks ECDLP in polynomial time on a CRQC; Regev-style methods may yield asymptotic improvements under stated assumptions, though practical superiority depends on constants, architectures, and instance structure.[1, 9, 10]

6.3 Symmetric Ciphers and Hashes

In principle, Grover motivates doubling key length to offset the quadratic speed-up; in practice, parallelization limits and error-correction costs determine wall-clock feasibility. Conservative guidance for long-lived data favors AES-256 for confidentiality and SHA-384/512 for integrity.[23, 24]

6.4 Analysis of Quantum Vulnerability

Quantum algorithms impact classical cryptosystems unevenly: symmetric schemes lose strength quadratically under Grover’s algorithm, while number-theoretic systems fail outright once Shor-capable fault-tolerant machines exist. As shown in Table 2, adapted from foundational 2019 estimates [37], previous literature-reported benchmarks summarize the qubits and time required to break major cryptosystems.

Table 2: Literature-Reported Estimates of Quantum Resilience for Current Cryptosystems, adapted from *Quantum Computing: Progress and Prospects* (National Academies of Sciences, Engineering, and Medicine, 2019) [37].

Cryptosystem	Category	Key Size	Security Param.	Quantum Algorithm	Logical Qubits	Physical Qubits	Time Required	Replacement Strategy
AES-GCM	Symmetric encryption	128	128	Grover's algorithm	2,953	4.61×10^6	2.61×10^{12} yrs	—
		192	192		4,449	1.68×10^7	1.97×10^{22} yrs	
		256	256		6,681	3.36×10^7	2.29×10^{32} yrs	
RSA	Asymmetric encryption	1024	80	Shor's algorithm	2,050	8.05×10^6	3.58 hrs	Move to NIST PQC
		2048	112		4,098	8.56×10^6	28.63 hrs	
		4096	128		8,194	1.12×10^7	229 hrs	
ECC (Discrete-log)	Asymmetric encryption	256	128	Shor's algorithm	2,330	8.56×10^6	10.5 hrs	Move to NIST PQC
		384	192		3,484	9.05×10^6	37.67 hrs	
		521	256		4,719	1.13×10^6	55 hrs	
SHA-256	Bitcoin mining	N/A	72	Grover's algorithm	2,403	2.23×10^6	1.8×10^4 yrs	—
PBKDF2 (10k iter.)	Password hashing	N/A	66	Grover's algorithm	2,403	2.23×10^6	2.3×10^7 yrs	Avoid password-based auth

Notes:

- Resource estimates assume a fault-tolerant quantum architecture with logical gates operating at a frequency of 5 MHz.
- These figures precede recent optimizations in quantum algorithms and error correction codes documented in [7, 8].
- Latest analyses highlight significant overheads in Grover-based AES key recovery, suggesting that symmetric encryption remains practically secure against near-to-mid-term quantum threats.[23].

6.5 Framework for Quantum Risk: Mosca’s Theorem

The strategic risk posed by the algorithms analyzed in the previous sections is formally characterized by Mosca’s Theorem, defined by the inequality $x + y > z$. A security collapse occurs if the required secrecy duration (x) plus the migration time (y) exceeds the estimated time until a CRQC (z).^[15] This inequality is currently being driven toward failure by the accelerated hardware roadmaps of major industry leaders.

7 PQC and NIST Standards

7.1 Finalized PQC Standards (Aug 13, 2024)

- **FIPS 203 (ML-KEM):** Based on the CRYSTALS-Kyber submission, this Module-Lattice Key Encapsulation Mechanism (KEM) is the primary standard for general-purpose encryption. It utilizes the hardness of the Module Learning with Errors (MLWE) problem. NIST offers three parameter sets corresponding to AES-128, 192, and 256 security strengths (ML-KEM-512, -768, and -1024). It ensures robustness against adaptive chosen-ciphertext attacks, making it the “drop-in” replacement for RSA-KEM and Elliptic Curve Diffie-Hellman (ECDH).^[17]
- **FIPS 204 (ML-DSA):** Derived from CRYSTALS-Dilithium, this is the primary digital signature standard for general use (e.g., TLS handshakes, document signing). It relies on the dual hardness of MLWE and Module Short Integer Solution (MSIS) problems. It was selected for its balanced performance, offering a superior trade-off between signature size and computational speed compared to traditional RSA signatures.^[18]
- **FIPS 205 (SLH-DSA):** Derived from SPHINCS+, this standard utilizes stateless hash-based signatures. Unlike ML-KEM/DSA, SLH-DSA does not rely on lattice assumptions. It is a conservative fallback based solely on the security of cryptographic hash functions (e.g., SHA-256, SHAKE256). While it produces significantly larger signatures and requires higher computational overhead, it provides a crucial safety net should unforeseen vulnerabilities be discovered in lattice-based cryptography.^[19]

7.2 In-Development and Additional Selections

- **FIPS 206 (FN-DSA / Falcon):** Based on the Falcon algorithm, this standard utilizes NTRU-lattice signatures over the ring of algebraic integers. NTRU stands for “N-th Degree Truncated Polynomial Ring Units” (and is historically referred to as the “Number Theory Research Unit”). FN-DSA provides the most compact signature and public key sizes among lattice-based candidates. However, its implementation is significantly more complex, requiring fast-Fourier sampling and high-precision floating-point arithmetic. Now moving through final validation, FIPS 206 is intended for resource-constrained environments where bandwidth is the primary bottleneck.^[21]
- **Code-based KEMs:** NIST selected HQC (Hamming Quasi-Cyclic) in 2025 to diversify KEM families beyond lattices in the standardization pipeline. HQC serves as a non-lattice alternative to ML-KEM, relying on the hardness of decoding random linear codes (specifically, the Syndrome Decoding Problem).^[20]

Standard	Primitive	Notes
FIPS 203 (ML-KEM)	KEM (MLWE)	Kyber; three parameter sets; early protocol integration and testing in hybrids.
FIPS 204 (ML-DSA)	Signature (Module-SIS/MLWE)	Dilithium; general-purpose PQ signatures with strong implementation guidance.
FIPS 205 (SLH-DSA)	Signature (Hash-based)	SPHINCS+; large signatures but structurally conservative fallback.
FIPS 206 (FN-DSA)	Signature (NTRU-lattice)	Falcon-like; draft in progress with floating-point/validation caveats.
HQC (pipeline)	KEM (Code-based)	Diversifies beyond lattices; selection announced 2025.

Table 3: NIST PQC standards status (as of early 2026).

8 Results and Discussion

8.1 Risk Posture and Timelines

Recent reductions in qubit estimates for RSA-2048 factoring (from ~ 20 million to below one million, and potentially to $\sim 10^5$ under QLDPC assumptions) may reduce theoretical resource estimates for CRQC feasibility, yet practical realization remains uncertain.[7, 8] Given HNDL, long-lived data encrypted today with vulnerable public-key schemes may be at risk if not rekeyed to PQC.

Emerging hardware platforms demonstrating operation at elevated temperatures (e.g., electron qubits on solid neon) indicate progress toward more practical and scalable quantum systems. While such approaches do not yet achieve fault tolerance, they reduce key engineering barriers, thereby increasing uncertainty in long-term timelines.

8.2 Implementation Attacks: KyberSlash

The KyberSlash timing attacks (KyberSlash1/2) target secret-dependent division timings in some Kyber (ML-KEM) implementations (including reference code prior to late-2023 patches), demonstrating practical key recovery on Cortex-A7/M4 under repeated decapsulation. The scheme remains sound; the lesson is that *constant-time*, *constant-memory-access* implementations and rigorous validation are mandatory for PQC rollouts.[22]

8.3 Operational Guidance for Symmetric Primitives

Given Grover’s quadratic speedup for key search, the effective post-quantum security of a k -bit symmetric key is reduced to roughly $k/2$ bits. This motivates the use of AES-256 for long-lived confidentiality, as it maintains approximately 128 bits of post-quantum strength.

For hash-based constructions, it is important to distinguish preimage and collision security. Grover’s algorithm reduces preimage resistance from n bits to $2^{n/2}$ operations, but collision resistance is governed not by Grover but by the Brassard–Høyer–Tapp (BHT) quantum collision algorithm, which reduces the cost of collision search to approximately $2^{n/3}$; however, practical feasibility for BHT requires huge quantum memory (qRAM) and tight models. Consequently, shorter digests such as SHA-256 provide limited long-term collision security in the

quantum setting, whereas SHA-384/512 retain adequate margins. Consistent with NIST’s key-management recommendations, AES-256 and SHA-384/512 are preferred for long-lived post-quantum protection.[3]

Among contemporary hash constructions, SHA-3 and its SHAKE extendable-output variants are widely regarded as strong candidates for post-quantum applications due to their conservative security margins, flexible sponge-based design, and compatibility with several standardized PQC schemes. Notably, SHA-3 was standardized in FIPS 202 prior to the publication of the recent NIST PQC standards, and its SHAKE variants are already incorporated into multiple standardized post-quantum cryptographic constructions.[16]

8.4 From Quantum Criticality to Risk: Reformulating Mosca’s Inequality

Historically, the parameter z in Mosca’s Inequality has been estimated under surface-code-based assumptions, placing the physical qubit requirements for RSA-2048 factorization in the range of 10^7 – 10^8 . Recent research, however, suggests that this timeline may be compressible along three primary axes:

- **Algorithmic Refinement:** Alternative approaches, such as Oded Regev’s factoring algorithm, trade increased repetition for reduced circuit depth, lowering the asymptotic depth from $\tilde{O}(k^3)$ toward $\tilde{O}(k^{1.5})$. While the total computational cost remains comparable, this shift relaxes requirements on sustained coherent execution and may better align with architectures constrained by logical qubit lifetimes and parallelization limits.
- **Architectural Optimization:** Emerging fault-tolerant designs based on QLDPC codes indicate the possibility of substantial reductions in physical qubit overhead compared to surface-code approaches. These gains, however, remain contingent on demanding architectural assumptions, including non-local connectivity, high-throughput syndrome extraction, and fast classical decoding, and have not yet been validated at large scale.
- **Engineering Improvements:** Advances in qubit materials and device engineering further influence the parameter z . Improvements in coherence times, noise resilience, fabrication quality, and operating conditions may accelerate system scalability, thereby compressing the timeline to cryptanalytically relevant quantum capabilities under optimistic assumptions.

Nevertheless, the realization date of a CRQC remains highly uncertain, and estimates on the order of 10–20 years should be interpreted as assumption-dependent rather than predictive.

While Mosca’s Inequality, $x + y > z$, is traditionally interpreted as a binary threshold for cryptographic failure, this formulation obscures the continuous nature of exposure accumulation under the HNDL threat model. To address this limitation, the inequality can be reformulated as a signed temporal margin:

$$\tau = x + y - z$$

where τ represents the temporal distance from the critical boundary. The regime $\tau > 0$ corresponds to deterministic vulnerability, whereas $\tau \leq 0$ represents a conditionally secure regime in which CRQC-enabled compromise is not guaranteed.

However, realistic confidentiality models rarely exhibit abrupt value expiration. In practice, the value of encrypted information typically decays gradually over time rather than vanishing instantaneously after a fixed secrecy lifetime. Consequently, the regime $\tau \leq 0$ should be interpreted as representing negligible—rather than strictly zero—risk under non-ideal assumptions.

Furthermore, the accumulation of exposed ciphertext during the migration period introduces a temporal amplification effect: as migration is delayed, the quantity of vulnerable data systematically increases, thereby raising the potential impact of future quantum decryption capabilities.

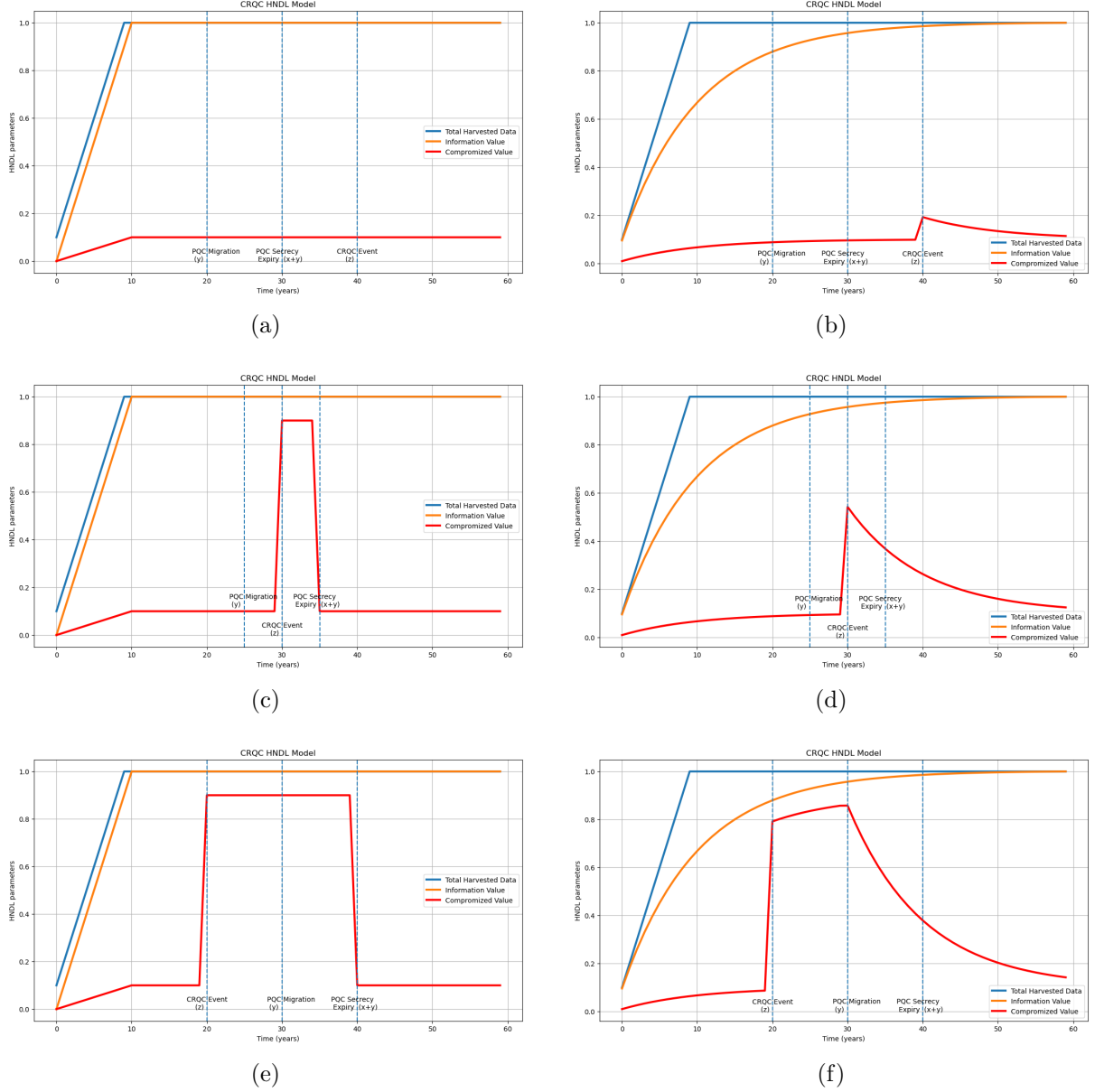


Figure 1: Illustrative HNDL exposure scenarios under different relationships between secrecy lifetime, migration duration, and CRQC realization timelines.

(a) Fixed-lifetime information-value model with secrecy lifetime $x = 10$ years satisfying the condition $x + y < z$, corresponding to a conditionally secure regime under Mosca's Inequality.

(b) Exponentially decaying information-value model with secrecy time constant $\tau = 10$ years. Residual information value persists beyond the nominal secrecy horizon, resulting in limited but non-zero HNDL exposure. In this regime, the binary interpretation of Mosca's Inequality becomes less precise due to continuous value decay.

(c) Partial overlap between the CRQC realization timeline and a fixed-lifetime secrecy model, producing limited but deterministic HNDL compromise.

(d) Partial overlap between the CRQC realization timeline and an exponentially decaying secrecy model, resulting in temporally distributed HNDL exposure.

(e) Deterministic vulnerability regime under a fixed-lifetime secrecy model where $x + y > z$, resulting in substantial HNDL compromise.

(f) Deterministic vulnerability regime under an exponentially decaying secrecy model, producing significant initial compromise followed by gradual exposure decay.

Figure 1 illustrates representative HNDL exposure scenarios under both fixed-lifetime and gradually decaying information-value assumptions. In these examples, adversaries initially accumulate encrypted data until reaching practical storage or resource constraints. Beyond this point, older low-value ciphertext may be discarded and replaced with newer, higher-value information, effectively maintaining a rolling HNDL exposure window.

Accordingly, post-quantum risk should be interpreted not as a binary condition, but as a continuous function of both the temporal margin τ and the dynamics of ciphertext accumulation. This perspective emphasizes that the urgency of migration is driven not only by uncertainty in CRQC realization timelines, but also by the deterministic growth of HNDL exposure as migration is delayed.

9 Conclusions

Recent developments in quantum cryptanalysis have placed classical public-key infrastructure on a definitive deprecation path. The high probability of Mosca’s Inequality failing—driven by the contraction of the threat timeline via emerging fault-tolerant pathways and QLDPC-based optimizations—underscores the immediate necessity of transitioning to NIST PQC standards (FIPS 203–205). While symmetric primitives such as AES-256 and SHA-512 retain adequate security margins through increased key and digest sizes, the vulnerability of current asymmetric systems requires an immediate shift toward cryptographic agility. By diversifying hardness assumptions through the integration of lattice, hash, and code-based primitives—facilitated by forthcoming standards like FIPS 206 (FN-DSA) and HQC—organizations can ensure long-term resilience against both known quantum algorithms and future cryptanalytic breakthroughs.

10 Recommendations

1. **Adopt standardized PQC now:** Prioritize ML-KEM (FIPS 203) for key establishment and ML-DSA (FIPS 204) for signatures; use SLH-DSA (FIPS 205) where conservative assurance is needed.
2. **Deploy hybrid modes:** Use hybrid (classical+PQC) in protocols (e.g., TLS 1.3) during transition for defense-in-depth.
3. **Harden implementations:** Enforce constant-time coding, side-channel testing, and validated libraries; ensure prompt patching (e.g., KyberSlash mitigations) and independent verification.
4. **Strengthen symmetric baselines:** Prefer AES-256 and SHA-384/512 for long-lived confidentiality/integrity; reevaluate key lifetimes under HNDL.
5. **Build crypto-agility:** Inventory cryptographic dependencies (CBOM), automate certificate lifecycle, and design for algorithm agility aligned with transition guidance.
6. **Monitor research:** Track resource-estimate updates (e.g., sub-million-qubit factoring), FIPS 206 progress, and code-based KEM maturation (HQC).

11 Challenges and Future Research

11.1 Technical and Implementation Challenges

- **Error-correction engineering:** Demonstrating fast, scalable QLDPC decoding and robust fault-tolerant operation remains a central challenge.[8]

- **Efficient PQC implementations:** Reducing footprint and energy on constrained devices while maintaining side-channel resistance is essential for practical deployments.[17, 18, 19]
- **Protocol integration:** Optimizing certificate chains, handshake sizes, and fallback strategies with ML-DSA/FN-DSA and ML-KEM/HQC to fit diverse operational environments.[21, 20]
- **Cryptanalysis:** Continued scrutiny of lattice assumptions, structured rings, and conservative alternatives (hash-based signatures; code-based KEMs) will increase confidence in deployed systems.[19, 20]

11.2 Mosca-Based Risk Models for HNDL Threats

As discussed previously, existing formulations of Mosca’s Inequality primarily define a threshold condition for cryptographic failure but do not fully capture the continuous accumulation of exposure associated with the HNDL threat model. Consequently, future research should investigate the development of quantitative risk models derived from Mosca’s framework that estimate the effective exposure of harvested information over time.

Such models should account not only for uncertainty in the realization timeline of CRQCs, but also for the temporal value of encrypted information, data accumulation dynamics during migration periods, and the persistence of long-term confidentiality requirements. In addition, these frameworks should incorporate both quantum-enabled decryption risks and classical non-quantum threats, thereby enabling a unified assessment of cryptographic exposure across hybrid adversarial environments.

A generalized Mosca-based risk framework may therefore provide a more realistic foundation for evaluating post-quantum migration urgency, HNDL impact severity, and long-term cryptographic resilience.

11.3 Foundational Mathematical Considerations

Modern cryptography—both classical and post-quantum—relies fundamentally on unproven computational hardness assumptions rather than formal mathematical certainties. As noted by Goldwasser and Kalai, the security of cryptographic constructions ultimately depends on conjectured intractability properties, such as the existence of one-way functions or the hardness of structured algebraic problems.[29] These assumptions, while extensively studied, lack definitive proof.

Consequently, cryptographic security should be understood as conditional: advances in number theory or computational complexity could, in principle, alter the hardness landscape on which existing schemes depend. However, such developments remain speculative and are not considered operational risks in current system design or post-quantum transition planning. This perspective applies uniformly across classical, symmetric, and post-quantum cryptographic primitives.

A related foundational property arises in cryptographic hash functions. By the Pigeonhole Principle, any fixed-length hash function must admit collisions, as it maps a larger input space into a smaller output space. This observation is consistent with principles from information theory, which preclude lossless compression of arbitrary data. Accordingly, hash function security derives not from collision avoidance, but from the computational difficulty of finding such collisions or preimages.

In both classical and quantum settings, this distinction remains essential: security guarantees are based on assumed computational limits rather than mathematical impossibility. As such, the long-term resilience of cryptographic systems depends on continued validation of these assumptions and, where appropriate, diversification across independent hardness foundations.

11.4 Lightweight Cryptography

A persistent challenge in constrained platforms (e.g., mobile, embedded systems, and industrial control systems) is the limited availability of FIPS-approved lightweight symmetric primitives. Current NIST guidance primarily standardizes block ciphers such as AES (and legacy TDEA for decryption), with no dedicated stream cipher approved for federal use.[30, 31] This creates a gap between standardized cryptography and the practical needs of resource-constrained environments.

In practice, several efficient and widely deployed alternatives exist outside the FIPS framework. For example, *ChaCha20*, a stream cipher designed by Bernstein, is standardized in the IETF as part of the ChaCha20-Poly1305 AEAD construction (RFC 8439) and provides 256-bit security with high performance on software platforms lacking AES acceleration.[33] Despite its widespread adoption (e.g., in TLS 1.3), it remains non-FIPS-approved.

More recently, NIST has addressed constrained environments through its lightweight cryptography initiative, selecting *Ascon* as the basis for its dedicated standard (SP 800-232). Ascon provides authenticated encryption and hashing with low computational overhead and is specifically designed for IoT and embedded systems.[32] However, it is not a stream cipher in the traditional sense, but rather a permutation-based AEAD scheme optimized for lightweight use cases.

Other lightweight block ciphers, such as *PRESENT* and *Speck*, further illustrate this landscape. PRESENT is an ultra-lightweight substitution–permutation network cipher with 64-bit blocks and 80/128-bit keys, designed for minimal hardware footprint.[35] Speck, an ARX-based block cipher family introduced by the NSA, is optimized for software efficiency across a wide range of block and key sizes.[36] While both have been studied extensively and deployed in niche contexts (e.g., RFID and IoT), neither is included in NIST’s core FIPS-approved set, and Speck in particular has faced controversy regarding standardization and trust.

Finally, recent FIPS publications related to PQC (FIPS 203–205) focus exclusively on asymmetric primitives, while the forthcoming FIPS 206 addresses digital signatures. None of these introduce new lightweight or stream cipher standards, reinforcing the observation that symmetric cryptography within FIPS remains centered on AES rather than newer lightweight or software-optimized designs.

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